Fiscal Year:	FY 2015	**Task Last Updated:**	FY 09/14/2015
PI Name:	Ferl, Robert Ph.D.		
Project Title:	Molecular Biology of Growth and Cell Remodeling within the Spaceflight Environment		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Gainesville	**State:**	FL
Zip Code:	32611-0690	**Congressional District:**	3
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2011 Space Biology NNH11ZTT002N
Start Date:	08/01/2012	**End Date:**	06/30/2016
No. of Post Docs:	3	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA KSC
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date is now 06/30/2016 per KSC information (Ed., 7/25/16) NOTE: End date is now 12/31/2015 per NSSC information; original end date 7/31/13 (Ed., 6/24/14) NOTE: End date is now 6/30/2014 per NSSC information; original end date 7/31/13 (Ed., 8/9/13)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Paul, Anna-Lisa (University of Florida)		
Grant/Contract No.:	NNX12AN69G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Plants experiencing spaceflight are quite normal in appearance but can exhibit growth habits distinctly different from plants on Earth. Historically the spaceflight-induced differences were difficult to dissect due to changing hardware and flight conditions. With the Advanced Biological Research System (ABRS) and a consistent International Space Station (ISS) orbital environment, it is now possible to confidently approach a dissection of the molecular biology guiding the altered growth of plants in spaceflight. This proposal involves experiments that specifically address growth and molecular changes in Arabidopsis that occur during spaceflight, bringing the molecular and genetic tools of Arabidopsis to bear on fundamental questions of root morphology, growth, and cell wall remodeling. During the Advanced Plant Experiments on Orbit - Transgenic Arabidopsis Gene Expression System (APEX-TAGES) experiments, we observed a remarkable number of gene expression changes during spaceflight that are associated with cell wall restructuring, especially in roots. We also observed, with the ABRS/GIS (GFP Imaging System) imaging system, genetically dependent changes in root growth direction, movement, and structure. The current experiments will employ the ABRS and the GIS to dissect and better understand the molecular changes guiding growth restructuring on orbit. In particular we hypothesize that spaceflight conditions engender a constant state of root cell wall modification during the cell file rotations that characterize spaceflight-induced root skewing. The specific experiments will involve mutants in root growth and cell wall signaling pathways identified in our previous spaceflight molecular studies, as well as new Green Fluorescent Protein biosensor lines also based on the genes identified in the previous flight experiments. The overall goal of our program is to understand fundamental molecular biological responses to extraterrestrial environments. This goal aligns with Fundamental Space Biology goals to understand the impact of spaceflight on biological systems to better enable the exploration imperative. We have used molecular biology and genetically tagged plants as biological monitors of spaceflight and space-related environments with great science return, leading to the development of new biological and hardware tools to study the spaceflight response with even higher fidelity in a wider variety of spaceflight-related environments. The focus of the current proposal is to build on these data and other insights gained from the execution of the APEX-TAGES experiments in the ABRS /GIS, advancing science while also advancing the telemetric imaging hardware and biological experiment support for the ISS and future platforms Gaining insights into such mechanisms is recognized as fundamental within the decadal study and underpins answers to some of the biggest questions in spaceflight plant biology.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	To gain fundamental insight into how plants perceive and process the signals that control tropism and adaptive physiology in novel and extreme environments. It also highlights how the unique research environment of the ISS can address biological questions that cannot be answered on Earth where gravity would mask many of the underlying phenomena.
Task Progress:	The major operations of the past year's reporting period was the Science Verification Test II, conducted in the VEGGIE hardware in June 2014, the Payload Verification Test, conducted in VEGGIE hardware, and finally the flight experiment of TAGES-ISA APEX03 on the International Space Station. The Science Verification test II (SVTii) was conducted in two parts – one at Kennedy Space Center (KSC) in the ISSES orbital environment simulator chamber, and a parallel study in the PIs' laboratory at the University of Florida (UF-SVTii). These plants were evaluated using the general morphology of the plants (growth patterns, health, and vigor). We used 12 day old plants as the primary developmental stage (and the only developmental stage in the KSC-SVTii), but would use information gathered in SVTii to reduce the age of plants in later experiments. The VEGGIE compatibility test demonstrated that although a substantially different growth platform than ABRS, it could be adapted to the plant support requirements defined by this proposal. The Payload Verification Test (PVT) was carried out following with the adjustments made to the experiment after the SVTii in VEGGIE. PVT was analyzed for quality of RNA that was recovered. The results from the PVT indicated that the experiment was prepared for launch. The final section of this report covers the preparations, launch and return of the APEX-03-02/TAGES-ISA experiment. Summary of planning and on orbit operations are included. All samples have been returned to the lab and are currently being processed and analyzed. I. SVTii setup and Operations SVT for APEX-03 was repeated (SVTii) to adjust to new hardware for the experiment. Due to the loss of ABRS/GIS unit onboard the ISS, accommodations were made to grow the APEX-03 plants in the VEGGIE unit. This caused minor changes to be made to the experimental design, including the ages of the plants and the way the plates are imaged. Plate planting: Arabidopsis seeds were planted on square, nutrient agar Petri plates. Two types of plates were prepared – an imaging plate to be inserted in to the Light Microscopy Module (LMM) and plates that would be harvested with one seed type per plate. The LMM plate would be used for imaging on flight and not harvested. Since the SVT had been run previously further molecular analysis was not conducted. Minor changes were made to the experimental design after the SVTii, included shortening the 5 and 12 day ages to 4 and 11 days to allow for consistent development morphology compared to plants grown previously in ABRS. This difference was caused by higher light levels in the VEGGIE hardware. II. PVT: Payload Verification Testing Using the same procedures outlined in the SVTii above, plates were prepared for the Payload Verification Test (PVT). Due to the change of hardware from the GIS to VEGGIE, a number of changes were made to the operations. The length of growth time for plants was reduced from 5 days to 4 days and from 12 to 11 days while 8 day old plates remained unchanged. Since VEGGIE does not have imaging capabilities built into the system, each set of plates (4 Day, 8 Day, and 11 Day) were photographed after being removed from VEGGIE and before being placed into the KFT. LMM: The Light Microscopy Module (LMM) on the ISS was to be utilized during flight in place of the GIS in the ABRS unit. The LMM imaging plate contained multiple GFP-transformed lines of Arabidopsis were used in order to observe protein localization in spaceflight.In order to provide ease of reference, small fluorescent beads were placed on the plate to help with imaging the plants. These beads ranged in size from 0.77microns to 100 microns. The PVT for the LMM was conducted separate from the experimental PVT at KSC. PVT for LMM was run at Glenn Research Center in

	Cleveland, Ohio. During the PVT, imaging was performed on all ages of plants that would be imaged in flight. Parameters that would be needed for flight operations were set and verified. III. Flight Experiment Flight operations were based in part on the lessons learned from the SVTs, and then tested in PVT operations. The launch and return of samples were successful, and the full report on the flight operations will be included in the final report for the grant to be submitted later in this year. Operation Objectives: The objectives were to expose 27 petri plates of plant seedlings to the ISS environment, take quality photograph of the plates, harvest the seedlings to RNAlater, and return the samples for post-flight analysis. Plates were harvested at different time points to help elucidate the different molecular responses of the varied developmental ages. In addition, a single plate from each age group was used for imaging of GFP reporter genes in the LMM. Growth Set-Up: Plates are activated on station by removing black wrapping. Plates are then placed in racks that are then placed in VEGGIE. VEGGIE has the ability to accommodate up to 30 plates at a time. However, due to a shared hardware chamber, not all the APEX-03 plates were run concurrently. The 11 day plates were grown first, followed by a second insertion of 8 and 4 day plates. Analyses of flight samples are currently on-going.
Bibliography Type:	Description: (Last Updated: 11/27/2022)
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